1.4 DNA and protein synthesis

i Know the structure of DNA, including the structure of the nucleotides (purines and pyrimidines), base pairing, the two sugar-phosphate backbones, phosphodiester bonds and hydrogen bonds.

ii Understand how DNA is replicated semi-conservatively, including the role of DNA helicase, polymerase and ligase.

iii Know that a gene is a sequence of bases on a DNA molecule coding for a sequence of amino acids in a polypeptide chain.

iv Know the structure of mRNA including nucleotides, the sugar phosphate backbone and the role of hydrogen bonds.

v Know the structure of tRNA, including nucleotides, the role of hydrogen bonds and the anticodon.

vi Understand the processes of transcription in the nucleus and translation at the ribosome, including the role of sense and anti-sense DNA, mRNA, tRNA and the ribosomes.

vii Understand the nature of the genetic code, including triplets coding for amino acids, start and stop codons, degenerate and non-overlapping nature, and that not all the genome codes for proteins.

viii Understand the term gene mutation as illustrated by base deletions, insertions and substitutions.

ix Understand the effect of point mutations on amino acid sequences, as illustrated by sickle cell anaemia in humans.